MISSION REPORT

Rømer I

Status:	PLANNING
Launch Date:	No launch date set
Operator:	ESA
Target:	Mun

This document contains technical specifications and trajectory plans for the Rømer I mission as part of the Kerbin Expansion Initiative.

May 15, 2026

Mission directory

Mission objectives

The primary goal of the mission is to test the launch vehicle, the cryogenic upperstage and the docking and life-support capabilities of the LEM (Lunar Exploratory Module) and of the LCM (Lunar Command Module).

The verification of the planned trajectory to the Mun, including the ascent and the trans-lunar injection is also of importance. The crew will be on a free-return trajectory, making sure that if anything will happen to the service module, like a failed engine ignition, the crew will still safely fall back down to Kerbin. For subsequent missions, the plan is to put the vehicle on a free-return trajectory, and after testing, the path will be modified to a prograde or retrograde (depending on fuel margins) orbit around the Mun. This mission will therefore also serve as practice for designing optimal free-returns.

Although landing on the Mun and the subsequent rendez-vous with the LCM are planned for Rømer II, testing the parameters of the mission and practicing the maneuvers needed for a successful landing and eventual return to Kerbin is paramount.

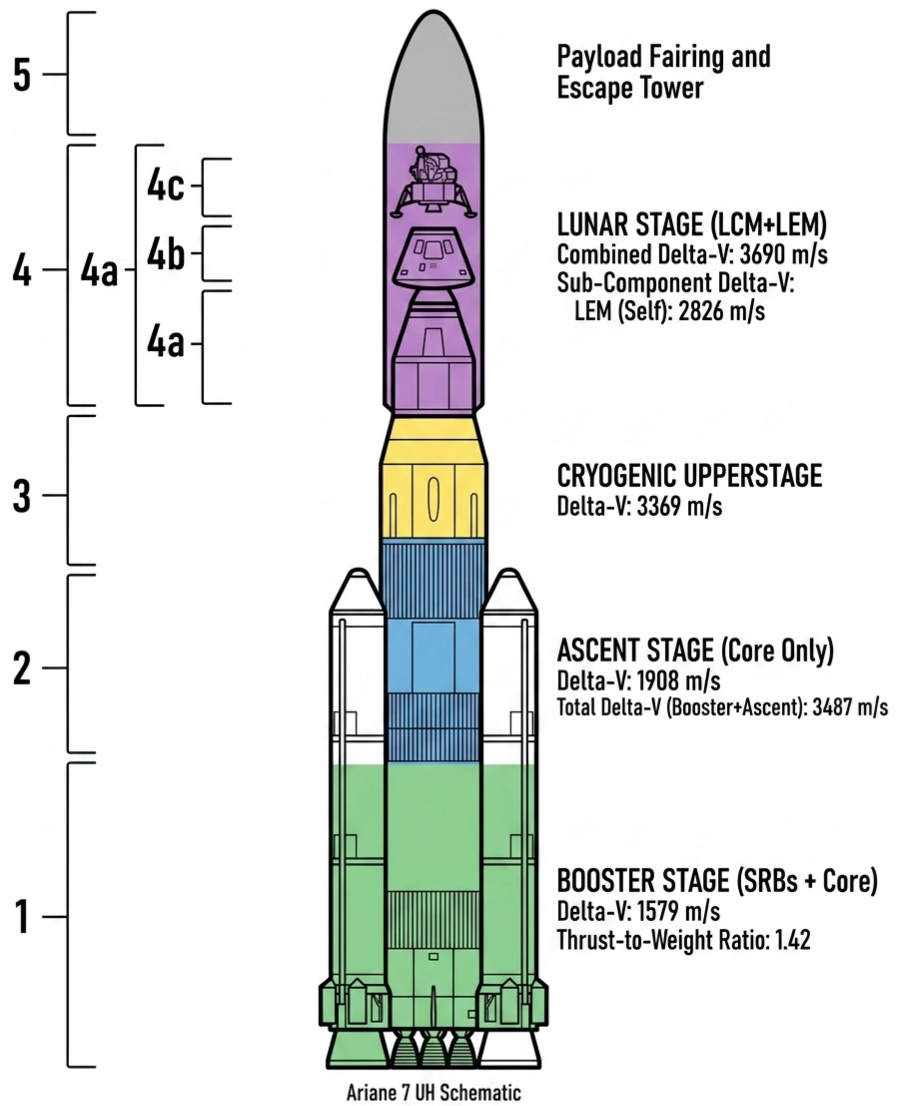

Figure 1: Schematic of the different stages and their associated performance numbers

Vehicle specifications

Presented next are the booster and the lander configurations used for the mission. Some information is excluded here for the sake of brevity but may be found in the description documents for the individual parts, namely the booster and the LEM.

2.1 Ariane 7 UH booster

It is a standard Ariane 7 UH booster. The core stages are kept intact. Even though the booster has addons for refueling in orbit and airbrakes for landing, they were not removed in preparation for the mission. The payload is the combined weight of the LEM, LCM, its decoupling infrastructure and the fairing used to protect them during ascent.

Below are the engineer readouts for the performance of the booster given the payload present:

Table 1: Ariane 7 UH Stage Specifications

Stage	**Body**	**Mass (t)**	**ISP (s)**	**Thrust (kN)**	**TWR (Max)**	**Δv (m/s)**
S3	Mun	43.243/71.977	350.0	250	5.06 (15.3)	3690
S4	Mun	130.892/202.869	340.0	2,000	6.05 (16.62)	3,369
S5	Kerbin (Vac)	294.839/497.708	305	4,050	0.83 (1.58)	1,919
S6	Kerbin (ASL)	695.27/1,192.98	230.9	15,646.6	1.34 (2.73)	1,615

A total of 10703 m/s (excluding the lander) was the total for this payload. The capabilities of the lander were excluded from the table (stages S0 through S2).

Stage S6 is the combined power of the first stage and the SRBs. It is comprised of 3 liquid fuel engines (called the core stage) and 4 boosters. Its role is to get the rocket through the thickest part of the atmosphere. Without the SRBs, it is doubtful that the 3 liquid fuel engines alone would have the TWR high enough to do that efficiently. Even though the engines are optimized for sea-level pressure, their efficiency rises in the upper parts of the atmosphere, increasing the Δv.

Once the side boosters are spent, the radial decouplers will detach. That is stage S5. At that point, a good amount of fuel is left in the core stage, but it is high enough in the atmosphere to be able to carry the rest of the rocket on its own. Along with the radial decouplers, some side-mounted separating boosters will push the SRBs safely away from the main rocket. Now it's just the core stage until orbit.

After an expenditure of about 3600 m/s, the core stage is now empty and it's detached as well. We enter stage S4, the cryogenic upperstage. It is so called because it uses liquid oxygen and liquid hydrogen as oxydizer and fuel respectively. This will give us the last 200 m/s necessarry to achieve a stable LKO (Low Kerbin Orbit). It would also have enough fuel for the TLI, but that is withing the purview of the LCM. The upperstage is equipped with remote control computers. After separation, we deorbit the stage to remove space clutter and debris. Optionally, if we choose to make an apoapsis raise to conserver fuel before the TLI, this upperstage could be used for that and discarded.

Finally, stage S4 is used for the TLI. Since we are planning a free-return trajectory, no other burns except for course correction are needed.

Figure 2: Picture of the LEM during construction

2.2 Lunar Exploratory Module (LEM)

The LEM is a light vehicle equipped with landing legs and enough fuel to land from a circular low lunar orbit and achieve orbit from the ground to rendez-vous with the LCM.

It is equipped with 4 Mk-1H Torch liquid engines and RCS capabilities. The RCS was added so that the crew will be able to control its attitude and make fine adjustments when docking with the LCM.

The MEL has no storage capacities, two solar pannel arrays and 150 units of electricity. Electricity was not of major concern, since the lander will never be piloted remotely. Of bigger concern was the amount of monopropellant. Along with the built-in monopropellant reserve next to the crew module, there are two more tanks of 7.5 units attached radially.

It also presents with a Clamp-O-Tron docking port on the forward side and LT-2 landing legs on the aft side. The landing legs are detachable to save weight on the ascent from the Mun, but the reliability of the decoupler is questionable and needs further study.

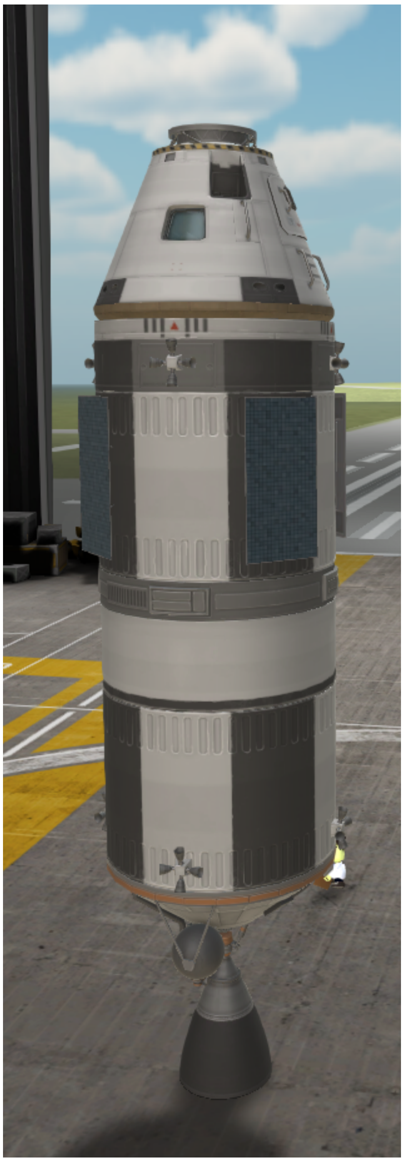

Figure 3: Picture of the LCM during construction

2.3 Lunar Command Module (LCM)

The LCM is where the crew will stay for the entirety of the mission, except for some tests when in proximity with the moon. It is the main operating grounds for maneuvers, communications and has the most life-support. Unlike the LEM, there is more space and even a toilet in the capsule.

The European Service Module is a subpart of the LCM, and is responsible for controlling the engines, the RCS and the SAS systems. After return to LKO and while on a trajectory to intercept the atmosphere, the ESM will be discarded and it will safely burn up in the atmosphere.

It has two deployable solar panel arrays and enough Δv to carry the LEM and the crew to the Mun and back.

Crew

As always, the crew requires at least a pilot and an engineer. A scientist is preferred but not needed. A second pilot is also a possibility. The crew will be limited to 3 people because of the Mk1-3 command pod used. Althought technically there are 2 such command pods on the rocket assembly, one is for the lander and one is for the command module. Upon docking during the trans-lunar trip, two kerbals (a pilot and engineer) will transfer to the LEM to prepare it for the landing procedure while the third crew member will stay in orbit with LCM.

List of the chosen team members:

- Hadbal Kerman — engineer and mission specialist
- Kerzer Kerman — pilot and LEM commander
- Malong Kerman — pilot and LCM commander

Trajectory

The trajectory poses multiple challenges from a mathematical standpoint, chief of which is getting the correct injection angle into the Mun's SOI. The ascent and the TLI itself are commonly done maneuvers.

Additionally, fuel-saving measures will be taken to maximize efficiency, although not strictly needed since the launch vehicle (unmodified, which is not the case here since we removed stage S3) can deliver approximately 305 tonnes to TLI, would be a good test to make for future missions to optimize for fuel. These optimizations will come in the form of splitting up expensive burns in multiple parts, as needed.

4.1 Ascent to LKO

The ascent will be faily standard. Immediate after clearing the launch tower, a roll program will be initiated to orient the rocket towards heading 90°. The roll should finish by the time the rocket hits 100 m/s going straight up. After that, a nominal pitch maneuever will start executing to angle the rocket further east, at about 20° per minute. The actual pitch speed will depend on factors like drag and time to apoapsis. It is preferable that the time to apoapsis stays at about 1 minute throughout the gravity turn.

After 1 minute and 26 seconds, the SRBs will have finished their fuel and will be detached from the main booster.

Nominal height at this point in the flight is approximately 20 km.

The core booster will continue burning for another 2 minutes and 54 seconds at full throttle. The actual time will depend on throttle settings to keep the apoapsis time to 1 minute.

The desired parking orbit before other maneuvers is a circular equatorial orbit of 75 kilometers. At that altitude, orbital speed is given by the formula:

$$v = \sqrt{\mu\left(\frac{2}{75+600km} - \frac{1}{75+600km}\right)}$$

$$v = 2287m/s$$

At an orbital speed of about 2.1 km/s, the core stage is separated. This ensures the core stage will remain on a suborbital trajectory and burn off in the atmosphere instead of being left in a stable orbit. The remaining 100 m/s for circularization can be achieved with the cryogenic upper stage.

Assuming optimal gravity, drag, and steering losses, total Δv_{ascent} should be about 3250 m/s, but we should assume 3800 m/s in the worst-case scenario.

4.2 Apoapsis raise

This is a fuel-saving technique and can therefore be omitted for the TLI instead, but the exact phase angle and Δv calculations will be different.

The purpose of this maneuver is to take advantage of the Oberth effect when raising the apoapsis for the Hohmann transfer to the Mun. Because the orbit is circular and equatorial, the burn has the same fuel requirements anywhere, but it should be 180° from where the desired apoapsis should be. The target orbit is an orbit with a period of exactly 1 Kerbin sidereal day: 5h 59m 9.425s or 21549.425s. The periapsis will necessarily remain at 75km. The apoapsis is therefore found by calculating the semi-major axis of the target orbit:

$$T_1 = 2pi \cdot \sqrt{\frac{a_1^3}{\mu}}$$

$$a_1 = \sqrt[3]{\frac{\mu \cdot T_1^2}{4\pi^2}} = \sqrt[3]{3.5316 \cdot 10^{12} \cdot \frac{21549.425^2}{4\pi^2}}$$

$$a = 3\,463\,334\,\mathrm{m}$$

Therefore, with a periapsis of 75 km, this results in an apoapsis of:

$$r_{a,1} = 2a_1 - r_{p,1} = 6251.668\,\mathrm{km}$$

$$ap_1 = r_{kerbin} - r_{a,1} = 5651.668\,\mathrm{km}$$

At periapsis, at this new target orbit, the speed should be:

$$v_{periapsis,1} = 3073.15\,\mathrm{m/s}$$

Subtracting the previous value of 2287 m/s, this results in a prograde burn of 786 m/s to raise the apoapsis.

The exact true anomaly at which we need to perform this burn will depend on the location of the Mun. Since this is the first burn in a two-burn maneuver that will result in the TLI, we need to make sure that the Mun will be at the correct place when we perform our TLI exactly one sidereal day from now.

4.3 Trans-lunar injection (TLI)

Let the period of the first orbit be $T_1 = 21\,549\,\mathrm{s}$ and the period of the second orbit be T_2 with a semi-major axis a_2, a periapsis of 75 km and an apoapsis of ap_2. The radius of the apoapsis will then be $r_{ap,2} = r_{kerbin} + ap_2$. A good starting value for this radius would be the distance between the centre of mass of Kerbin and the Mun plus the sphere of influence of the Mun. So:

$$r_{pe,2} = r_{kerbin} + 75\,\mathrm{km} = 675\,\mathrm{km}$$

$$r_{ap,2} = r_{kerbin} + d_{kerbin,mun} + r_{SOI,mun} = 600 \cdot 10^3 + 12000 \cdot 10^3 + 2429.559 \cdot 10^3 = 15\,029\,000\,\mathrm{m}$$

$$a_2 = 7\,852\,279.5\,\mathrm{m}$$

$$T_2 = 2\pi\sqrt{\frac{a^3}{\mu}} = 73\,568\,\mathrm{s}$$

To achieve this orbit from the previous T1 orbit, vis-viva gives us a speed of:

$$v_{periapsis,2} = \sqrt{\mu(\frac{2}{r_{pe,2}} - \frac{1}{a_2})} = 3164.52\,\mathrm{m/s}$$

Resulting in a prograde burn of 91.34 m/s.

Calculating the approximate phase angle between the burn the Mun will require finding out where the Mun was $T_1 + T_2/2$ seconds before our encounter. Luckily, the Mun has a perfectly circular orbit, so there's no need converting back and forth from time to mean anomaly, since at an eccentricity of 0, true anomaly is equal to mean anomaly. However, in order to compare the angles of the ship and the Mun, both must be converted to true latitudes. Luckily, this is a simple addition.

For a circular orbit, the angular velocity (in radians per second) of the Mun is given by:

$$\omega_{mun} = \frac{2\pi}{T_{mun}} = 4.520\,784 \times 10^{-5}\,\text{rad/s}$$

But

$$T_{total} = T_1 + T_2/2 = 21549 + 73568/2 = 58\,333\,\text{s}$$

Taking the angular velocity formula and rearranging to find the angle the Mun makes after a certain Δt:

$$\theta = \omega_{mun} \cdot \Delta t = 2.6371\,\text{rad} = 151.09°$$

Therefore $\theta - 180°$ is the phase angle at which the vessel needs to do the first burn (the burn responsible for the T1 orbit) such that after the TLI, the new apoapsis will coincide with the exact location of the Mun. In reality, this is a rough approximation, as the ship needs to encounter the Mun on the leading side of its orbit around Kerbin, so subtracting around 10° from this value would place the ship in the approximate correct location for a free-return trajectory. The exact angles needed would require studying the impact parameter, the flight path angle and doing a state vector simulation to be exact.

This angle is not the difference between the true anomalies of the two objects. True anomalies are calculated based on the location of the periapsis (the argument of periapsis or ω) and the latitude of the ascending node (LAN or Ω). As such, after getting reading for all these values, calculating the true longitude for each is trivial:

$$L_{ship} = \Omega_{ship} + \omega_{ship} + \nu_{ship}$$

$$L_{mun} = \Omega_{mun} + \omega_{mun} + \nu_{mun}$$

When the value of L_{ship} is $\theta - 180°$ behind the Mun is the correct place to do the first burn.

One needs to take care that the value wraps from 360° to 0° during the calculations for the ship's true longitude, since it takes more than one orbit for the ship to complete the T_1 and half of the T_2 orbit to get to the Mun.

Subtle corrections can be made from this orbit to get a free-return trajectory that puts the ship's periapsis around Kerbin after the Mun encounter into the atmosphere at about 20–30 km.

4.4 Free-return

This phase of the flight is from the completion of the TLI burn until the encounter with Kerbin's atmosphere, approximately 2 days after the TLI. At the beginning of this phase is also when the separation of the upper stage and the docking maneuver is supposed to be tested. The upper stage can then be maneuevered remotely to place it on an impact trajectory to the Mun.

Subtle corrections may be needed in this phase to correct for the inclination or for slight errors during the docking sequence of the LEM and LCM. Although we started with the assumption that the T_0 orbit around Kerbin is equatorial and every subsequent burn was purely prograde, that may not be true. Although this doesn't change the calculations in the previous steps at all, for orbital hygiene purposes, inclination corrections may be done as long as one remembers to correct for the prograde-retrograde speed side-effects of such maneuevers.

Launch windows

There are 3 conditions for the launch:

- The launch should be during the day at the KSC launchpad. This puts the possible launch times everyday between 4:13 (sunrise) until 1:10 the next day (sunset). There is a strong preferance for the morning since the rocket will launch due east and that way it will still be in the sun's light until it achieves orbit.
- At least some part of the time of flight between Kerbin and the Mun during TLI will need to be illuminated. This happens in any possible configuration, since there is no realistic TLI burn that will keep the ship eclipsed by either Kerbin or the Mun. This condition enables the docking and maneuvering tests to be done in the light.
- The perilune should happen at a time such that the far side of the Mun is completely illuminated. This requires a solar eclipse from the perspective of Kerbin. Technically, it's a new moon, but since both Kerbin and the Mun have an ecliptical inclination of exactly 0, they are the same thing.

To calculate possible launch times, we need to take into account the duration of every stage of the trajectory, get a time estimate for when a lunar eclipse will happen anywhere on Kerbin starting on Year 2, day 167 and then extrapolate possible launch times and splashdown times.

Eclipses happen regularly on Kerbin, at a frequency of about 7 days. Not taking into account whether they are visible from the KSC, the following eclipses are proposed as targets for the perilune time:

1. Year 2, day 174, 1:07:36
2. Year 2, day 180, 4:19:32
3. Year 2, day 187, 1:31:27
4. Year 2, day 193, 4:43:23
5. Year 2, day 200, 1:55:18
6. Year 2, day 206, 5:07:13
7. Year 2, day 213, 2:19:09
8. Year 2, day 219, 5:31:04
9. Year 2, day 226, 2:43:00
10. Year 2, day 232, 5:54:55
11. Year 2, day 239, 3:06:50
12. Year 2, day 246, 0:18:46

And these are the known approximate times for the known phases of the mission:

- From mark to stable orbit — 680 seconds
- Coast time before apogee raise — depends on the phase angle between the Mun and the rocket
- Coast time before TLI maneuver ≈ 1 sidereal day or 21549 seconds
- From TLI to perilune ≈ 5 hours and 20 minutes or 19080 seconds
- From perilune to splashdown ≈ 5 hours and 30 minutes or 19800 seconds